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PRIMARY RESEARCH PAPER

Eutrophication drives resource use efficiency variations in freshwater plankton communities across river–lake continuum in China's Poyang Lake Basin

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Abstract Resource use efficiency (RUE) is a crucial ecological indicator to assess nutrient uptake by phytoplankton and grazing by zooplankton. This study estimated the RUE of plankton communities and quantified the relative effects of nutrient levels and physical and chemical factors on RUE in the Fu River to Poyang Lake continuum, revealing significant spatiotemporal differences. Phytoplankton RUE was significantly higher in rivers than that in lakes, while zooplankton RUE was higher in lakes than in rivers. Additionally, RUE was higher during normal and wet seasons compared to dry season. Linear mixed-effects models revealed that eutrophication are the key drivers of phytoplankton RUE in both rivers and lakes, showing negative correlations with Trophic Level Index (TLI) in the river, and with phosphate and nitrate-nitrogen in the lake. Zooplankton RUE is mainly driven by discharge and nitrite-nitrogen in river and lake systems, respectively. These insights underscore the remarkable ecological flexibility of plankton and further offer in-depth valuable perspectives to enhance river and lake water quality, controlling nutrient pollution and enhancing ecological functions.

Keywords Resource use efficiency · Phytoplankton & zooplankton · Eutrophication · Physical and chemical factors · Spatiotemporal variation · Poyang Lake-Fu River ecosystem

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Introduction

As the cornerstone of the freshwater ecosystem food web, plankton communities play an important role in the material cycling and energy flow of aquatic ecosystem (Cardinale et al., 2002). Due to contributing primary and secondary productivity,

resource use efficiency (RUE) has been widely used by ecologists as an important metric for studying local-scale productivity in freshwater ecosystems. RUE refers to the efficiency with which a community utilizes limited resources, and is typically characterized by the biomass yield per unit of available resource (Hodapp et al., 2019). The metric not only reflects an organism's activities but also their interactions with other organisms and environment. Biodiversity is essential for maintaining the functionality of aquatic ecosystems. Research has shown that more diverse communities tend to exhibit higher productivity and nutrient retention (Cardinale et al., 2002). Hence, the beneficial impact of species richness on phytoplankton resource use efficiency (RUE) has been widely documented (Ptacnik et al., 2008; Filstrup et al., 2014a; Otero et al., 2020). However, the structure of plankton communities is not solely determined by species richness. Functional traits trade-offs are also shaped by environmental filtering and biotic interactions (Kneitel & Chase, 2004). Under specific environmental conditions, species with traits best suited for survival tend to be selected, which is a process that can alter community structure and, in turn, affect functional roles such as resource conversion efficiency (Grime, 2006).

In natural aquatic ecosystems, physical, chemical, and nutritional factors collectively shape the habitat conditions for plankton communities (Geng et al., 2022). For instance, water temperature can influence phytoplankton biomass by altering their photosynthesis rates. Under specific nutrient conditions, increased temperatures can stimulate phytoplankton growth and metabolic rate, thereby enhancing productivity (Kim et al., 2019). Moreover, the correlations between nutrient concentrations and phytoplankton resource use efficiency (RUE) are intricate and multifaceted, and further amplified by pronounced spatial heterogeneity along aquatic continua that is characterized by distinct hydrological forcing (e.g., flow velocity gradients), light regimes, and nutrient stoichiometry, all of which exert divergent selection pressures on phytoplankton community organization (Breton et al., 2017; Li et al., 2023). However, the extent to which these ecosystem-specific constraints modulate the trait-RUE relationships across the river–lake ecosystems remains uncharacterized. Elucidating these context-dependent mechanisms is essential for advancing our understanding of freshwater ecosystem functioning under varying environmental scenarios.

In this study, using Poyang Lake (PYL) and its inflowing river (Fu River, FR) as case study, we aim to investigate the determinants of resource use efficiency (RUE) in phytoplankton and zooplankton across river–lake continuum employing field sampling, laboratory analysis, and statistical approaches. We elucidate their correlations between environmental variables (including water temperature, pH, and nutrient levels) and both plankton community structure and RUE. Furthermore, this study compares the differences in driving factors between riverine and lacustrine environments, providing insights into the ecological processes playing in each habitat. We hypothesize that (1) plankton RUE varies significantly with the hydrological cycle and differs along the river–lake gradient; (2) phytoplankton and zooplankton RUE are strongly influenced by environmental factors, such as water levels, discharge, depth, and nutrients with distinct spatiotemporal variations; (3) there are some differences in the drivers of RUE for phytoplankton and zooplankton, with eutrophication having a greater impact on RUE than the physical environment.

Materials and methods

Study area

Poyang Lake (24°29′14″N to 30°04′43″N; 113°34′18″E to 118°28′56″E) is China's largest freshwater lake, which is situated in north of Jiangxi Province, (Fig. 1). The region experiences a typical subtropical monsoon climate, with monthly mean water temperatures ranging from 5.9 to 29.9 ℃ (averaged 18 ℃ annually). The average annual precipitation is approximately 1500 mm, while annual pan evaporation measures around 1100 mm, both primarily occurred during the summer months, especially in July. Poyang Lake basin is crucial for sustaining freshwater resources and fish stocks, flood control, climate regulation, and pollution degradation. It is fed by five major rivers including Gan, Fu, Xin, Rao, and Xiu, and connects freely with the Yangtze River at its outlet. From April to June, the influx from these rivers increases dramatically, causing the lake's area to expand seasonally. During peak high-water levels

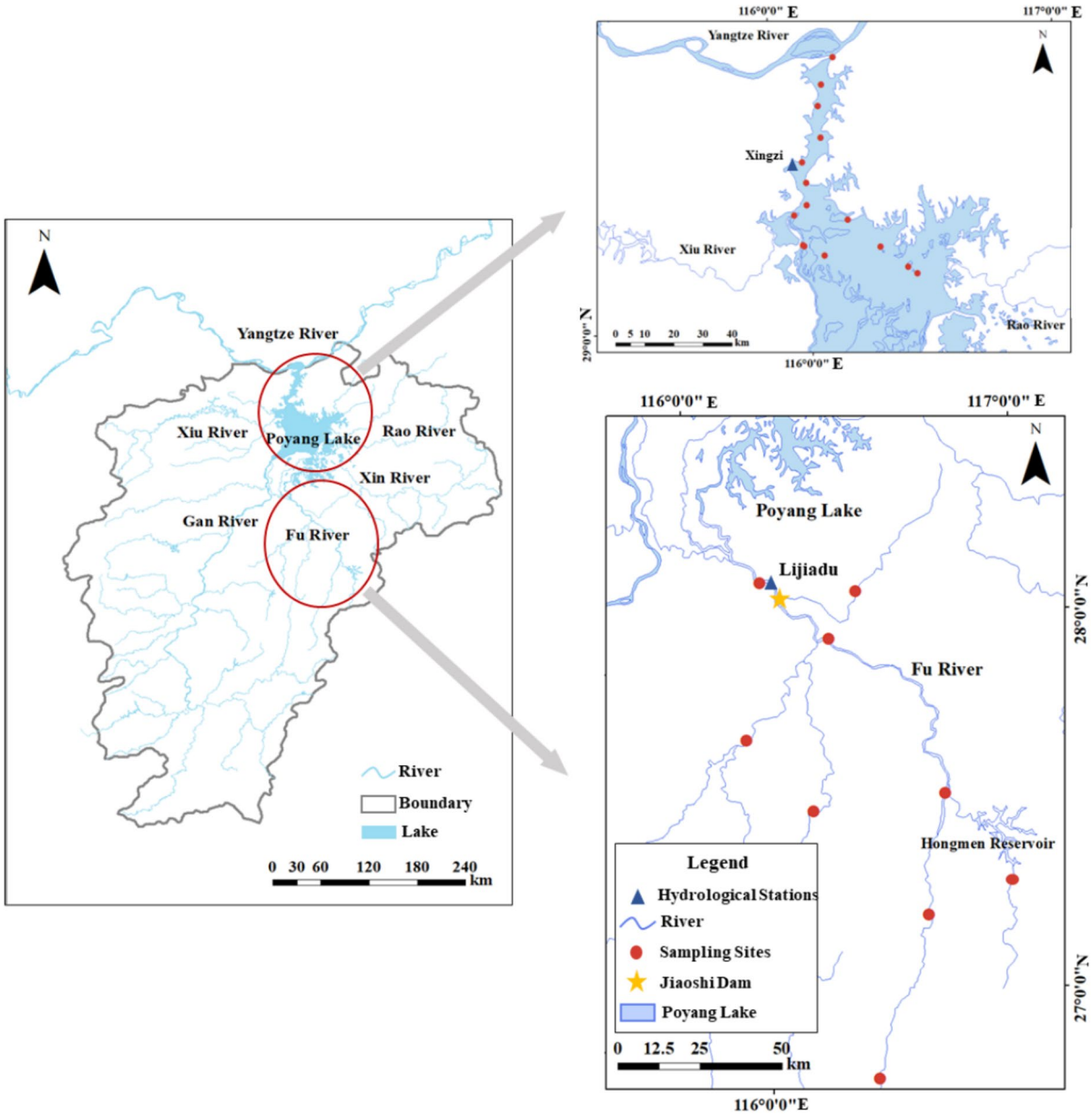

Fig. 1 Location of Poyang Lake and the Fu River, with sampling sites in the study area

(20 m, Wusong Datum), the lake's area can exceed 4125 km^2. Conversely, the lake's area can drop to below 1000 km^2 when water level fall to 10 m. The seasonal rising and falling of water levels cause significant variability in the area, depth, and connectivity of floodplain lakes across different seasons (Hurd et al., 2016; Li et al., 2020b). The Fu River, a major tributary of Poyang Lake, and the second largest river in Jiangxi Province, stretches 349 km in length with a basin area of 17,186 km^2. The river exhibits a similar annual flood pulse pattern, which divides the hydrological cycle into three distinct phases: normal season (from April to June, October and November, and the monthly average water level reaches a mid-value of the year), wet season (from July to September, with higher water levels and increased discharge, and during this season, floodwaters inundate floodplains, facilitating the transfer of matter and energy between

terrestrial and aquatic ecosystems. The saucer-shape lakes merge with the main lake, forming a cohesive water body), dry season (from December to March, the phase is marked by low water level. The hydrological connection between the rivers and lake area is disrupted, leading to the exposure of shoals and the fragmentation of algal habitats).

Samples collection and processing

We conducted field investigations in July and December 2017, and April 2018, corresponding to the wet, dry, and normal seasons, respectively. We established 9 representative sampling sites in both the mainstream and tributaries of the Fu River using a grid distribution method (Fig. 1). During these surveys, we collected samples the environmental parameters, phytoplankton, and zooplankton at each site.

Some indicators, including water temperature (WT), pH, specific conductance (SPC), and dissolved oxygen (DO) were assessed in situ with the portable water quality monitoring devices (YSI Professional Plus; YSI, Yellow Springs, OH, USA). Transparency was assessed using a Secchi disc (SD). After filtering each raw water sample through a 0.45 µm glass-fiber filter, we measured the chlorophyll a (Chl-a) spectrophotometrically. In addition, we collected 1-L water samples from near-surface depths (0.5 m) at each site and stored them in polyethylene bottles for measurement of total nitrogen (TN), total phosphorus (TP), nitrate-nitrogen (NO_3-N), ammonia nitrogen (NH_4-N), phosphate (PO_4-P), and chemical oxygen demand (COD) according to standard protocols (China Bureau of Environmental Protection, 2002).

Phytoplankton and zooplankton samples were collected from the surface waters (to a depth of 0.5 m) with a Plexiglass water collector. We collected 1-L water samples in polyethylene bottles, which were then fixed with 10–15 mL of Lugol's reagent. After standing for 48 h, the samples were condensed to 50 ml. For qualitative and quantitative analysis of zooplankton, we pre-filtered 20-L water samples through a 64 -µm mesh size plankton net, concentrated them into 50 ml, and fixed them with a 10% formaldehyde solution. Taxonomic identification and enumeration of species were conducted under a microscope (Olympus, CX21 400×, Germany) using methodologies outlined in the Methods of Freshwater Plankton Research (Zhang & Huang, 1995), and references such as the Freshwater Plankton Atlas (Hang, 1980) and Common Freshwater Planktonic Algae Atlas of China (Weng & Xu, 2010). All organisms were classified to the taxonomic levels of class, genus, or species. We used the geometry shapes approximating the actual cell shape to calculate the average cell biovolume, which we then converted into wet weight to estimate plankton biomass.

Despite seasonal fluctuations in water levels, the 15 sampling sites established in the Poyang Lake remained submerged year-round (Table S2). The dataset presented herein was derived from surveys conducted by the Poyang Lake Wetland Observation and Research Station under the Chinese Academy of Sciences between January, April, July, and October from 2013 to 2018 (Liu et al., 2020a) and was supplemented with data referenced from Qian et al. (2016) and Yang (2024).

A long-term hydrological dataset was obtained from the Yangtze River Basin hydrological yearbook (Changjiang Water Resources Commission, 2020), with the Xingzi and Lijiadu stations representing Poyang Lake and Fu River, respectively. We analyzed the water levels variations of Poyang Lake and discharges of Fu River from 1988 to 2022.

Plankton resource use efficiency

Plankton resource use efficiency (RUE) generally indicates the biomass yield per unit of available nitrogen or phosphorus. Total phosphorus (TP) and total nitrogen (TN) frequently employed used as metrics to assess the potential productivity of an aquatic ecosystem, thereby reflecting the availability of resource (Ptacnik et al., 2008; Hodapp et al., 2019; Chai et al., 2020). Referring to the definition provided by Jin et al. (2023), In this study, we have characterized phytoplankton resource use efficiency as the proportion of phytoplankton biomass to nutrient concentrations, as detailed below:

$$RUE_{pn} = \ln\left(\frac{Bio(phyto)}{TN} + 1\right) \quad (1)$$

$$RUE_{pp} = \ln\left(\frac{Bio(phyto)}{TP} + 1\right) \quad (2)$$

where $Bio(Phyto)$ represents phytoplankton biomass; TN is total nitrogen; and TP is total phosphorus.

Zooplankton resource use efficiency (RUE_{zp}) is determined by the fraction of phytoplankton biomass that is assimilated and transformed into zooplankton biomass:

$$RUE_{zp} = \ln\left(\frac{Bio(zoo)}{Bio(phyto)} + 1\right) \quad (3)$$

where $Bio(Zoo)$ denotes zooplankton biomass.

Data analysis

We employed generalized additive mixed models (GAMMs) to analyze the temporal dynamics of hydrological regimes. The model incorporated a smooth nonlinear trend over time using a thin-plate regression spline (s(YearMonth, bs = "tp")), along with a random-effects smooth (s(Season, bs = "re")) to account for seasonal variations. To address potential temporal autocorrelation in the residuals, an ARMA (autoregressive moving average) correlation structure was incorporated, modeling dependency within-year dependence to ensure residuals approximated a normal distribution. This flexible framework enabled the detection of complex, non-monotonic temporal patterns in hydrological processes. This analysis was performed by using R version 4.2.2 (R Core Team, 2022) with the *mgcv* package (Wood, 2017).

By applying a weight-distribution approach (Table S1), TN, TP, Chl-a, and SD were integrated into a unified index, namely the Trophic Level Index (TLI) (Wu et al., 2024), which is used to quantitatively characterize the trophic status of water bodies (Text S1). To assess whether the fluctuations in plankton density, and biomass and their resource conversion efficiency were correlated to collection seasons and locations, we employed Analysis of Variance (ANOVA) and Non-parametric tests in SPSS statistics 27.

To examine the effects of environmental factors on resource use efficiency (RUE), we implemented linear mixed-effects models (LMMs) using the *lme* function from the "*nlme*" package and multi-model inference via the "*MuMIn*" package in R (Pinheiro et al., 1999). The environmental factors except pH were ln-transformed before analysis. The models included a random intercept structured by hydrological seasons to account for temporal heterogeneity. The full model contains all environmental factors that have no significant correlation. The optimal fixed-effect structure was selected via multi-model inference, comparing all possible subsets using *AICc* (corrected Akaike Information Criterion) to ensure valid likelihood comparisons and compare the goodness of fit of LMMs. Models with a $\Delta AICc < 2$ were retained and ranked by their relative importance to identify the key factors exerting a significant impact on the response variable. To simplify the analysis, we excluded the interaction between factors. This multi-model inference and weight sorting allowed us to determine the critical factors affecting plankton resource utilization efficiency. The general structure of the model was as follow:

$$lmm(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_i x_i + Season_j + \epsilon_{ij}$$

here, y is the expected response of the RUE (RUE_{pp}, RUE_{pn}, and RUE_{zp}), β_0 is a constant, x_i is environmental variables, β_i is the fixed-effect coefficient of environmental variables, $Season_j$ is the random intercept in different seasons, ϵ_{ij} is residual.

The correlation of environmental factors was determined by Spearman correlation coefficients using the *corrplot* (Wei & Simko, 2024) and *dplyr* (Wickham et al., 2023) packages. All variables were Min–Max normalized before analysis. Factors with $|r| > 0.65$ was classified as highly correlated, with one representative retained for inclusion in the LMMs.

Results

Variations in hydrological regimes, physical and chemical characteristics, and plankton communities

Although the precipitation has remained relatively stable, the water level exhibits a downward trend (Fig. S1). The water levels of Poyang Lake exhibit a large variation range from 4 to 21 m annually. (Fig. S1b). The mean annual water level is 11.25 ± 0.03 m. The GAMM results show that the water level of Poyang Lake has been decreasing slowly year by year (edf = 1, $F = 6.252$, $p = 0.887$), and significantly influenced by seasons (normal, wet, and dry) (edf = 1.99, $F = 155.132$, $p < 0.001$) From 1988 to 2022, the river's water level exhibited a distinct pattern of oscillation with an overall downward

trend (Fig. S1e). Prior to 1991, there was a gradual decline, followed by an increase, peaking in 1997 at an annual average of 26.18 ± 0.17 m. After that, the water level experienced a continuous decline, albeit at a slower rate, with some recovery after 2010. However, between 2016 and 2022, the rate of decline accelerated. Due to the significant daily variation in flow, a logarithmic transformation was applied to improve the graphical representation. The flow of Fu River closely mirrored the direction of the water level change (either rising or falling) (Fig. S1h), but it revealed a more pronounced cyclical variation.

The variations in hydrological regimes exhibited pronounced seasonal differences ($p < 0.001$) (Table S3). We observed that the water levels of Poyang Lake exhibited in a stepwise increase, coinciding with the seasonal shift from the dry, normal, and wet. Water levels during these three seasons were significantly different ($p < 0.001$) (Fig. S1c). Similarly, in the Fu River, significantly higher water levels were recorded during the wet season (25.01 ± 0.20 m), which was significantly higher than that in dry season (Fig. S1f). Compared to the dry season (212.70.50 ± 22.90 m^3/s), the discharge during the normal and wet seasons increased by 100.46 m^3/s and 424.83 m^3/s, respectively.

Physical and chemical conditions are fundamental in shaping ecological communities, and seasonal changes significantly influence them in both the Fu River and Poyang Lake (shown in Fig. 2, Fig. S2). While parameters like dissolved oxygen (DO) and chemical oxygen demand (COD) exhibited no significant spatial variation, nutrient-related factors, such as total phosphorus (TP) and total nitrogen (TN), showed marked spatial heterogeneity. For instance, TN concentrations in the Fu River (1.49 ± 0.20 mg/L) were approximately three times higher than those in Poyang Lake (0.26 ± 0.10 mg/L). TP concentrations in Poyang Lake (2.72 ± 0.22 mg/L) exceeded those in the Fu River (1.32 ± 0.36 mg/L). The TLI calculated from them (Text S1), ranging from 41.44 to 62.14, and 34.65 to 67.86 in the Fu River and Poyang Lake, respectively, which was between mesotrophic and moderately eutrophic. Nitrate-nitrogen (NO_3-N) levels were higher in Poyang Lake (1.19 ± 0.06 mg/L) compared to the Fu River (0.83 ± 0.09 mg/L). The concentrations of phosphate (PO_4-P) (FR: 0.04 ± 0.01 mg/L; PYL: 0.04 ± 0.002 mg/L) and ammonia nitrogen (NH_4-N) (FR: 0.37 ± 0.09 mg/L, PYL: 0.38 ± 0.07 mg/L) were comparable between the two zones.

The variation in phytoplankton biomass exhibited extremely significant differences between the two zones, and the river consistently demonstrating higher values than that in the lake (Fig. 3a). In the Fu River, phytoplankton biomass peaked at 57.33 ± 6.86 mg/L during the wet season, compared to 47.87 ± 4.14 mg/L during the normal season, and 7.75 ± 1.11 mg/L during the dry season. Community composition exhibits distinct patterns across the three seasons.

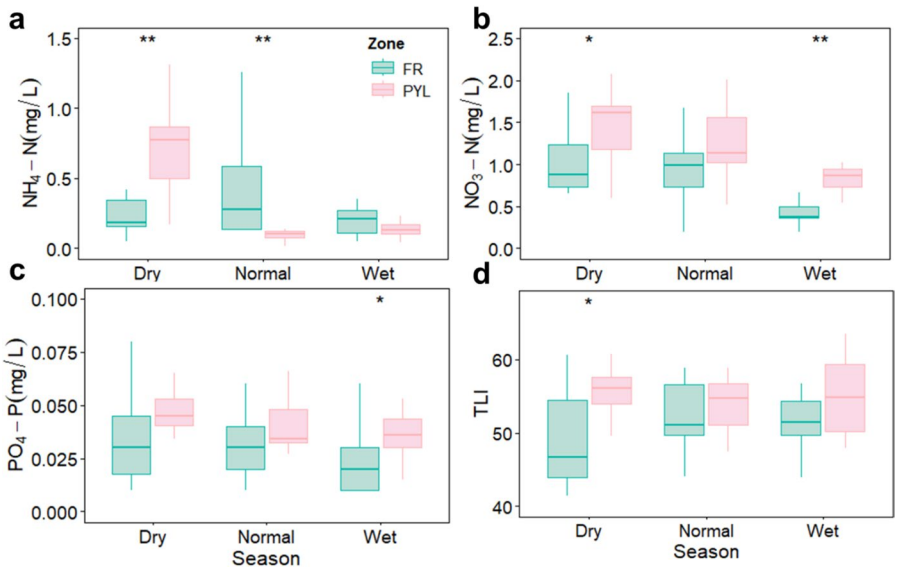

Fig. 2 Variation of physical and chemical parameters in the Fu River (FR) and Poyang Lake (PYL). Asterisks indicate significant differences between seasons, with $*p < 0.05$, $**p < 0.01$

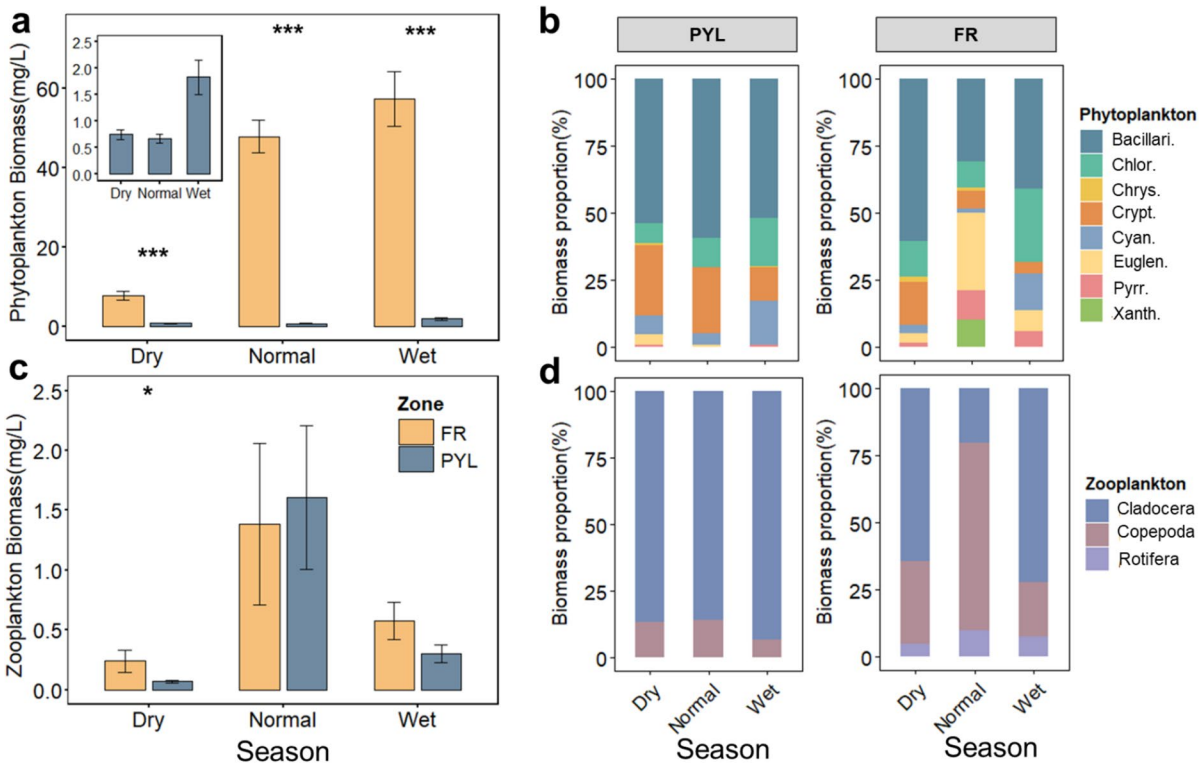

Fig. 3 Seasonal variations in biomass of phytoplankton and zooplankton and their composition in the Fu River (FR) and Poyang Lake (PYL). **A** Phytoplankton biomass, **B** Phytoplankton composition, **C** Zooplankton biomass, **D** Zooplankton composition; Bacillari.: Bacillariophyta, Chlor.: Chlorophyta, Chrys.: Chrysophyta, Crypt.: Cryptophyta, Cyan.: Cyanophyta, Euglen.: Euglenophyta, Pyrr.: Pyrrophyta, Xanth.: Xanthophyta. Asterisks indicate significant differences between zones, with $*p<0.05$, and $***p<0.001$

Notably, Bacillariophyta were the most abundant phylum, followed by Cryptophyta (Dry), Euglenophyta (Normal), and Chlorophyta (Wet), which accounted for 15.79%, 28.80%, and 27.43% of total community biomass, respectively. Additionally, Xanthophyta exhibited a minor but significant presence restricted to the normal season. In Poyang Lake, biomass ranged from 0.16 to 5.71 mg/L, with an average of 1.09 ± 0.14 mg/L. The wet seasons has significantly higher biomass levels (1.84 ± 0.33 mg/L) compared to the normal season (0.67 ± 0.12 mg/L) and the dry seasons (0.75 ± 0.12 mg/L; Fig. 3c). Bacillariophyta constituted the biomass predominant phylum in Poyang Lake, accounting for over 50% of the algal community (Fig. 3c). Nevertheless, Cryptophyta emerged as the second largest group during both the normal (24.68%) and dry seasons (26.32%). During the wet season, Chlorophyta and Cyanobacteria exhibited comparable abundances, comprising 17.89% and 16.05% of the community, respectively. Additionally, minor populations of Euglenophyta and Pyrrophyta (each < 10%) were detected (Fig. 3b).

Zooplankton biomass exhibited significant differences between the Fu River and Poyang Lake exclusively during the dry season ($p<0.05$). Zooplankton biomass of Poyang Lake (1.98 ± 0.79 mg/L) was higher than that of the Fu River (1.38 ± 0.68 mg/L) in normal season. While during the wet season, the biomass declined to 0.30 ± 0.07 mg/L in Poyang Lake and 0.57 ± 0.16 mg/L in the Fu River, reaching its lowest levels during the dry seasons (Fig. 3c). The zooplankton community in Poyang Lake was dominated by Cladocera and Copepoda, with Cladocera comprising over 85% of the total biomass. In the Fu River, the zooplankton assemblage included an additional Rotifera component, though this group remained minor (9%). Seasonal shifts in dominance were evident: Cladocera prevailed in both dry

(64.44%) and wet (72.30%) seasons, while Copepoda became dominant in the normal season (70.01%) (Fig. 3d).

Spatiotemporal variations of plankton RUE along the river–lake gradient

We observed significant shifts in the resource use efficiency of plankton along the river–lake gradient (Fig. 4). Specifically, both RUE_{pp} and RUE_{pn} significantly decreased from the Fu River to Poyang Lake ($p<0.001$). However, no significant differences in RUE were observed among the sample sites within each area, including the river (RUE_{pp}: $F=0.51$, $p=0.83$; RUE_{pn}: $F=0.62$, $p=0.74$; RUE_{zp}: $F=1.68$, $p=0.17$) and the lake (RUE_{pp}: $F=0.34$, $p=0.98$; RUE_{pn}: $F=0.37$, $p=0.97$; RUE_{zp}: $F=0.99$, $p=0.49$) (Fig. S4). Phytoplankton were more efficient at acquiring and transferring phosphorus than nitrogen. In the Fu River, RUE_{pp} declined progressively

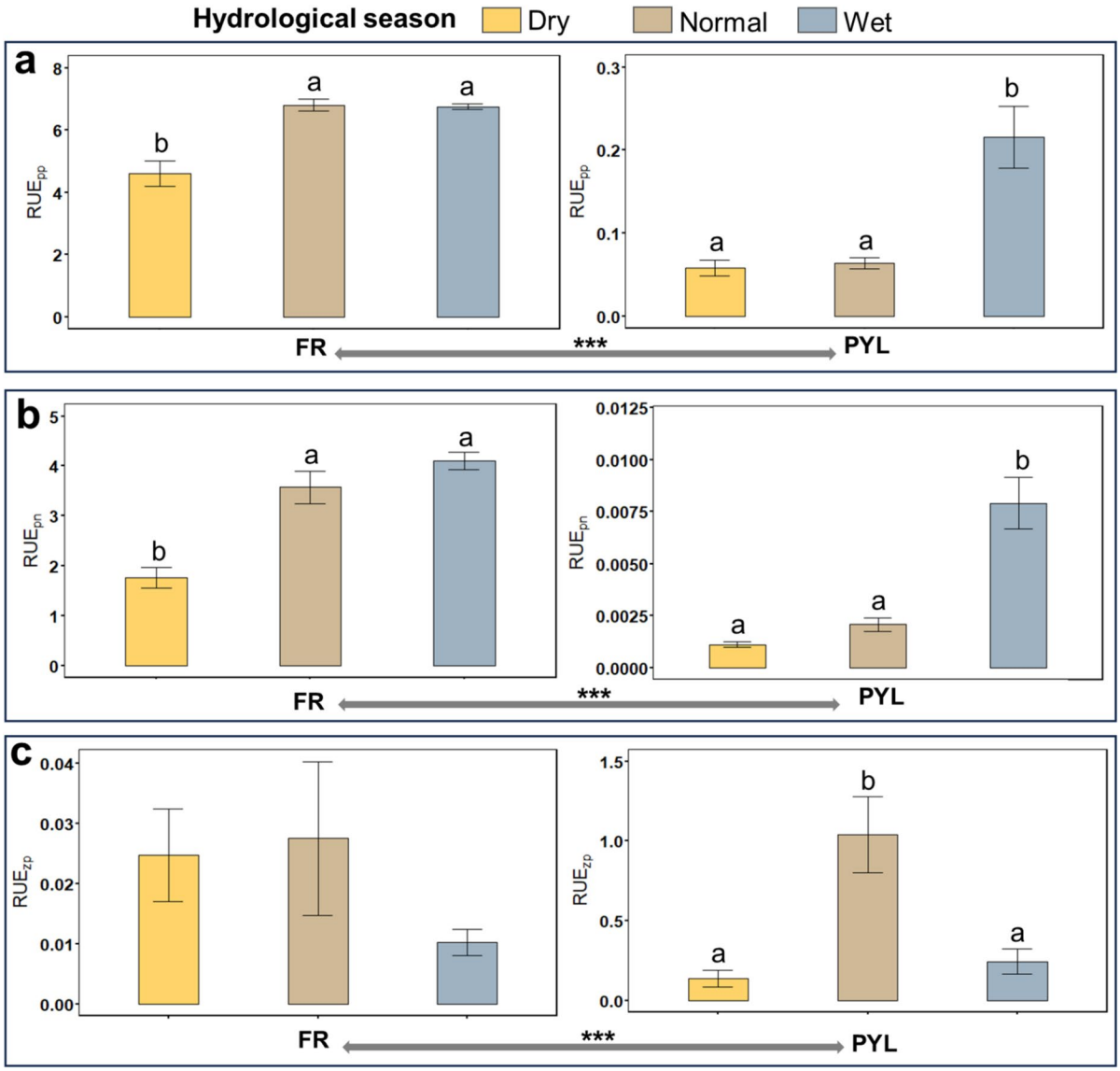

Fig. 4 Plankton resource use efficiency in the Fu River and Poyang Lake across three hydrological seasons. **A**, **B** above bars indicate significant differences among hydrological seasons; asterisks denote highly significance between the two zones, $p<0.001$

from the normal to the wet and dry seasons. The values were similar between the normal (6.80 ± 0.20) and wet (6.75 ± 0.10) seasons, but significantly lower in the dry season (4.59 ± 0.40; $p < 0.01$). Conversely, in Poyang Lake, RUE_{pp} increased across hydrological seasons: dry season (0.058 ± 0.038), normal season (0.063 ± 0.025), and wet season (0.215 ± 0.145). The variation of RUE_{pn} followed the same trend both in the Fu River and Poyang Lake, peaking in the wet season (FR: 4.10 ± 0.18, and PYL: 0.0079 ± 0.0012), followed by the normal season (FR: 3.57 ± 0.32, and PYL: 0.0021 ± 0.0003), and the lowest in the dry seasons (FR: 1.76 ± 0.20, and PYL: 0.0011 ± 0.0001).

However, the zooplankton resource use efficiency in Poyang Lake was markedly higher nearly 20 times than that in the Fu River (Fig. 4c). In both zones, the RUE_{zp} peaked during the normal season and reached its nadir during in the wet season. However, it did not exhibit significant variation across the three seasons in the Fu River. In contrast, in Poyang Lake, RUE_{zp} during normal season (1.04 ± 0.24) was approximately fivefold higher than that during the wet season (0.24 ± 0.08), and nearly sevenfold higher than that during the dry season (0.14 ± 0.05). Furthermore, the RUE_{zp} within the lake exceeded the RUE_{pp} and RUE_{pn}. Conversely, in the Fu River, RUE_{zp} was much lower compared to RUE_{pp} and RUE_{pn}.

Driving factors for variation of plankton resource use efficiency

Correlation analysis of environmental factors revealed distinct patterns between the two ecosystems. In the Fu River system, we observed highly significant ($p < 0.001$) negative correlations between: COD and DO ($r = -0.70$), SPC and water level (WL) ($r = -0.69$), and TLI and WL ($r = -0.71$), while SPC and TLI showed strong positive correlation ($r = 0.67$). Water temperature (WT) demonstrated moderate negative correlations with NO_3-N ($r = -0.50$, $p = 0.008$) and PO_4-P ($r = -0.43$, $p = 0.025$), but positive correlations with pH ($r = 0.43$, $p = 0.025$) and discharge ($r = 0.40$, $p = 0.012$). All other pairwise correlations were weak ($|r| < 0.43$) (Fig. 5a).

In Poyang Lake, WT showed particularly strong relationships with SPC ($r = -0.81$) and oxidation–reduction potential (ORP) ($r = -0.89$) (both $p < 0.001$), while SPC and ORP were also strongly correlated ($r = 0.82$, $p < 0.001$). pH displayed moderate negative correlations with COD and WT ($r = -0.58$), but positive correlations with SPC, ORP ($r = 0.52$), and NH_4-N ($r = 0.61$) (all $p < 0.001$). WT maintained moderate correlations with NO_3-N, NH_4-N, and water depth. Notably, SS and TLI showed minimal correlations with other parameters ($|r| < 0.25$). Nitrite-nitrogen (NO_2-N) exhibited no significant relationships except for a weak positive correlation with PO_4-P (Fig. 5b).

The model selection process identified potential optimal models ($\Delta AIC < 2$) for predicting resource use efficiency (RUE) across both ecosystems (Table 1). In the Fu River, RUE_{pp} was best predicted by TLI and water temperature (WT) as significant predictors ($p < 0.01$). RUE_{pn} was most effectively explained by a three-factor model (TLI + WT + PO_4-P; weight = 0.722) accounting for 86.7% of variance ($p < 0.05$ for all predictors), while RUE_{zp} was uniquely predicted by discharge (DIS;

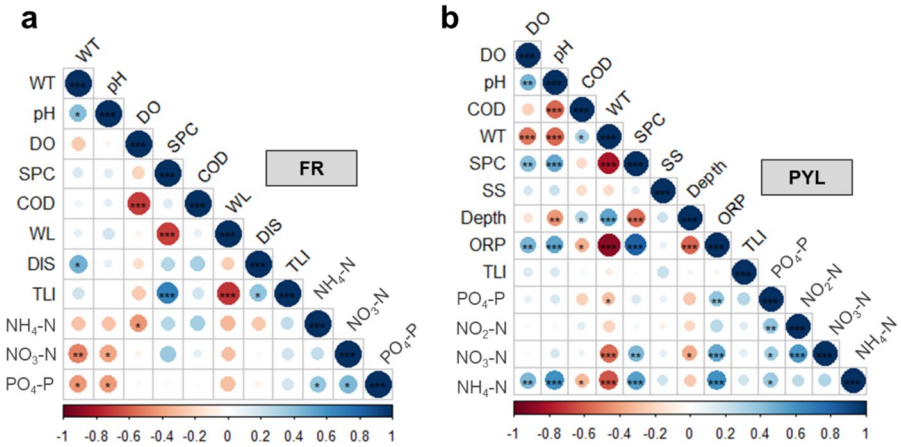

Fig. 5 Spearman correlation of environmental factors in the Fu River (**A**) and Poyang Lake (**B**); $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$

Table 1 Potential optimal candidates LMMs for predicting the effects of environmental factors on resource use efficiency (RUE)

Zone	RUE	Model	df	Log-lik	AIC_c	ΔAIC	Weight	R_m^2	R_c^2
FR	RUE_{pp}	$TLI^{***} + WT^{**}$	5	−15.825	45.3	0	0.508	0.772	0.907
		$TLI^{***} + WT^{**} + COD$	6	−15.555	46.6	1.24	0.273	0.818	0.906
		$TLI^{***} + WT^{**} + pH$	6	−15.572	47	1.68	0.219	0.758	0.907
	RUE_{pn}	$TLI^{***} + WT^{***} + PO_4\text{-}P^{*}$	6	−17.849	46	0	0.722	0.867	0.867
		$TLI^{***} + WT^{***} + PO_4\text{-}P^{*} + pH$	7	−17.497	47.9	1.9	0.278	0.871	0.871
	RUE_{zp}	DIS^{**}	4	56.318	−122.1	0	0.528	0.333	0.362
		$DIS^{**} + NH_4\text{-}N$	5	52.768	−120.5	1.59	0.239	0.364	0.44
		$DIS^{**} + pH$	5	53.406	−120.4	1.64	0.233	0.351	0.384
PYL	RUE_{pp}	$PO_4\text{-}P^{***} + TLI$	5	44.707	−90.4	0	0.467	0.224	0.561
		$PO_4\text{-}P^{***}$	4	44.436	−89.5	0.97	0.287	0.178	0.552
		$PO_4\text{-}P^{***} + TLI + COD$	6	42.714	−89.1	1.28	0.246	0.244	0.569
	RUE_{pn}	$NO_3\text{-}N^{*} + TLI$	5	186.222	−393.4	0	0.503	0.19	0.588
		$NO_3\text{-}N^{*} + TLI + pH$	6	180.412	−392.2	1.25	0.269	0.149	0.687
		$NO_3\text{-}N^{*} + TLI + NO2$	6	180.199	−391.8	1.58	0.228	0.204	0.587
	RUE_{zp}	$NO_2\text{-}N^{*} + NH_4\text{-}N + TLI$	6	−37.706	83.6	0	0.314	0.287	0.442
		$NO_2\text{-}N^{*} + NH_4\text{-}N$	5	−39.793	84.4	0.85	0.205	0.222	0.427
		$NO_2\text{-}N^{*} + NH_4\text{-}N + TLI + SS$	7	−38.196	84.7	1.16	0.176	0.304	0.458
		$NO_2\text{-}N^{*} + NH_4\text{-}N + SS$	6	−39.986	84.8	1.26	0.168	0.252	0.449
		$NO_2\text{-}N^{*} + NH_4\text{-}N + TLI + pH$	7	−38.596	85.3	1.68	0.136	0.338	0.499

Log-lik: loglikhood, is logarithmic measurement of model goodness-of-fit; Weight is the probability that the model is relatively optimal among all the candidate models; R_m^2 and R_c^2 are the marginal and conditional goodness of fit, respectively; * $p<0.05$, ** $p<0.01$, and *** $p<0.001$

weight = 0.528). In Poyang Lake, optimal predictive models included PO_4-P and TLI for RUE_{pp} (explaining 22.4% of variance), NO_3-N and TLI for RUE_{pn} (19.0%), and NO_2-N + NH_4-N + TLI for RUE_{zp} (28.7%).

To quantify the contribution of predictor variables, variable importance weights were used. These weights reflect both the inclusion probability in candidate models and the magnitude of significant effects to assess the relative contribution. Predictors with weight > 0.8 were identified as major determinants. The analysis revealed divergent yet comparable driving patterns across ecosystems: in the Fu River, RUE_{pp} was predominantly influenced by TLI (weight = 1) and WT (weight = 0.95) (Fig. 6a), while RUE_{pn} was jointly driven by WT, TLI, and PO_4-P (Fig. 6b). Notably, DIS emerged as the overwhelmingly dominant predictor for RUE_{zp} (weight = 0.98), surpassing all other factors (Fig. 6c). Contrastingly, Poyang Lake exhibited distinct key drivers, with PO_4-P, NO_3-N, and NO_2-N identified as the primary environmental determinants for RUE_{pp}, RUE_{pn}, and RUE_{zp}, respectively (Fig. 6d–f). TLI also approached a weight of 0.8.

Analyses of key factor impacts on RUE across different ecosystems highlighted eutrophication as a significant driver of phytoplankton RUE. In the Fu River, the Trophic Level Index (TLI) was the dominant control for RUE_{pp} ($\beta = -5.438$, $p < 0.001$) and RUE_{pn} ($\beta = -4.977$, $p < 0.001$). Conversely, in Poyang Lake, RUE_{pp} exhibited a significant negative correlation with PO_4-P ($\beta = -0.146$, $p < 0.001$) (Fig. 7d), while RUE_{pn} was negatively influenced by NO_3-N ($\beta = -0.004$, $p = 0.013$). Unlike the Fu River, TLI showed a positive but non-significant association with phytoplankton RUE in Poyang Lake (Table S5).

RUE_{zp} exhibited complex regulatory patterns: in the Fu River, it showed a negative correlation with discharge (DIS) (Fig. 7c). Conversely, in Poyang Lake, RUE_{zp} was positively influenced by nitrite-nitrogen (NO_2-N, $\beta = 0.296$, $p = 0.04$) though TLI ($\beta = -0.667$, $p = 0.42$) and ammonium (NH_4-N, $\beta = -0.148$, $p = 0.29$) exerted non-significant negative effects. These results highlight divergent nutrient

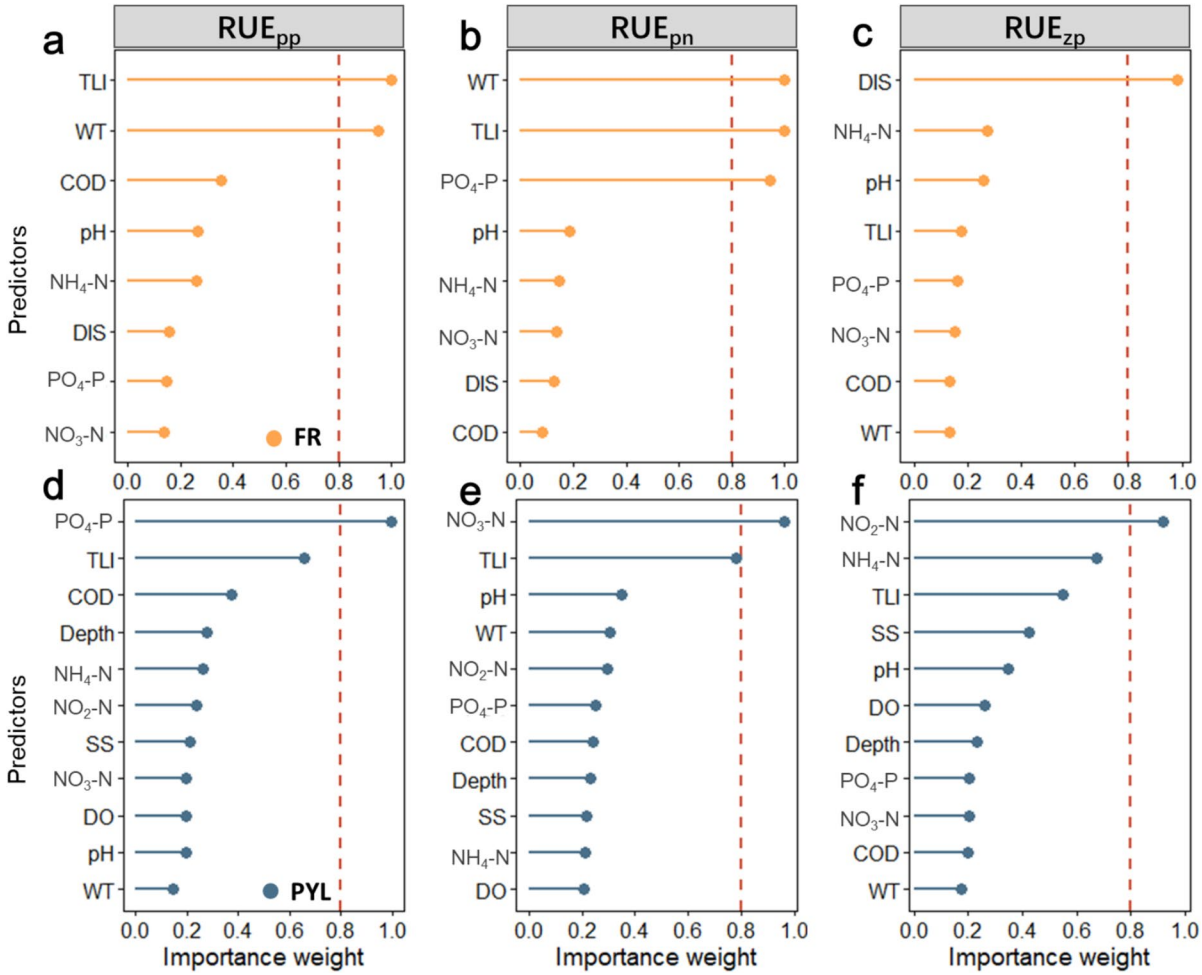

Fig. 6 Summary of the weights of each predictor in multi-model inference of RUE in the Fu River **A–C** and Poyang Lake **D–F** with the red dotted line indicating a weight threshold of 0.8

limitation patterns between the riverine and lacustrine ecosystems.

Discussion

Differences of resource use efficiency between rivers and lakes

Our results confirm our initial hypothesis, revealing significant differences in resource use efficiency (RUE) of plankton between rivers and lakes. Phytoplankton RUE was significantly higher in rivers than in lakes, both for phosphorus and nitrogen. The spatial differences of phytoplankton RUE have been reported in previous studies, for instance, Zhang et al. (2024a, b) reported substantial differences between RUE_{pp} and RUE_{pn} across four diverse aquatic ecosystems, including natural rivers, urban rivers, wetland, and lakes, they also demonstrated that RUE_{pn} in rivers was significantly higher in the lake, however, the RUE_{pp} was highest in the lake. Drowning from previous research, the increase of TP concentration may promote the growth of dominant species such as cyanobacteria, whereas the dominant species might not necessarily be the most efficient in utilizing TP, and the dominance of a single species will affect biodiversity and stability, thereby reducing RUE (Amorim & Moura, 2021; Geng et al., 2022). Our results showed that the TP concentration in Poyang

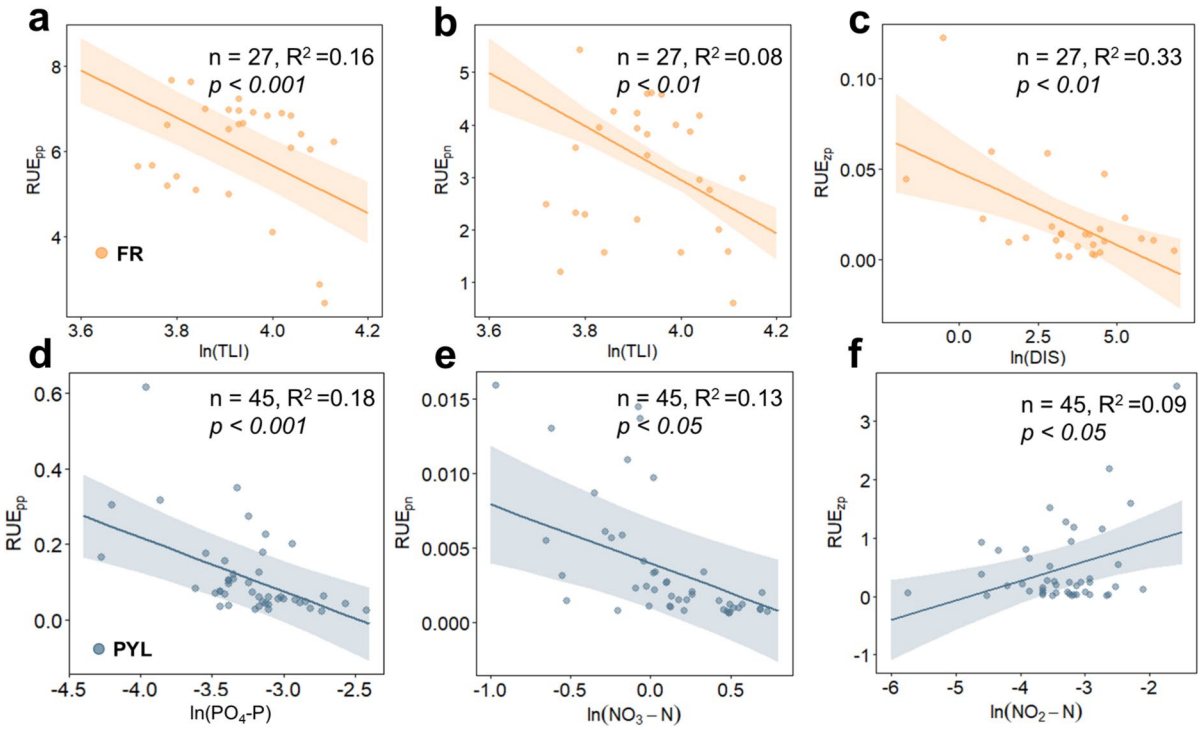

Fig. 7 Relationships between the key drivers and RUE_{pp}, RUE_{pn}, and RUE_{zp} in the Fu River **A**–**C** and Poyang Lake **D**–**F** n indicates the number of samples used in the models; Solid lines denote the regression lines, and colored shaded areas represent 95% confidence intervals

Lake increased significantly compared with that in the Fu River (Fig. S2h). In the environment with elevated TP concentrations, phytoplankton might experience a condition of nutrition saturation. We hypothesize that the high TP levels in Poyang Lake triggered a rapid proliferation of phytoplankton like cyanobacteria (Fig. S3b). However, this excess nutrient supply may not have been fully utilized, thereby contributing to the observed decline of RUE_{pp}.

However, zooplankton RUE (RUE_{zp}) was higher in Poyang Lake than in the Fu River. Straile (1997) points out that the stability of water bodies (e.g., lakes vs rivers) has a significant impact on zooplankton resource use efficiency, that lakes generally have more stable hydrological conditions and richer nutrient resources, which helps zooplankton use resources more efficiently, and zooplankton growth efficiency is higher at low food concentrations and decreased at high food concentrations. Food quality is directly related to zooplankton resource consumption. When Cyanobacteria, especially *Microcystis*, are dominant (Table S4), they have cross-trophic cascade effects on RUE_{zp} (Filstrup et al., 2014a). The bottom-up influence on zooplankton can be explained by two primary factors. Firstly, the substantial biomass of phytoplankton, which often consists of inedible or low-nutritional—quality forms, such as colonies or filamentous Cyanobacteria, thereby rendering zooplankton grazing less effective (Heathcote et al., 2016). Secondly, the increase in turbidity due to algal blooms, which outcompetes aquatic vegetation for light, thereby reducing available refuges for zooplankton from visually oriented predators like fish and other aquatic invertebrates (Amorim & Moura, 2021). This explains the higher phytoplankton biomass and the lower efficiency of zooplankton resource utilization in the Fu River.

Earlier research has demonstrated that the traits of regional habitats can profoundly influence the dynamics between nutrients and primary producers, particularly for phytoplankton, over extensive spatial extents (Filstrup et al., 2014b; Fergus et al., 2016). Rivers and lakes represent distinct freshwater habitats, with lakes characterized by relatively static water

bodies and rivers by their clear directional flow. The high mobility of river flows results in more frequent changes in the water environment, facilitating the dilution of pollutants and downstream migration of resources. These differences lead to distinct variations in water environmental conditions between rivers and lakes, driving spatial differentiation in the structure and functional traits within plankton communities. In Poyang Lake, Cyanobacteria and Chlorophyta are dominant phytoplankton phyla, while Bacillariophyta and Chlorophyta prevail in the Fu River (Fig. S3b). These algal groups above are ubiquitous in aquatic ecosystems, and play crucial roles in the nitrogen and phosphorus cycle (Hu et al., 2023). However, their preferences for nitrogen and phosphorus vary, and are influenced by environment factors, such as trophic state and temperature (Amorim & Moura, 2021; Li et al., 2024; Zhang et al., 2024a, b). Studies examining the breakpoints in RUE_{pn} along trophic gradients in lakes have shown that the shift from oligotrophic to eutrophic conditions markedly diminishes the efficiency with which nutrients are transferred from phytoplankton to zooplankton (Jin et al., 2023). Field observations in the Pearl River estuary have shown that the phytoplankton RUE driven by eutrophication is higher in the condition of low nitrogen and high phosphorus, and lower in the condition of high nitrogen and low phosphorus (Li et al., 2024). It is widely recognized that relatively static lakes are more susceptible to phosphorus enrichment than flowing rivers (Smith et al., 1999). Consistent with this, our findings confirm that the relatively static Poyang Lake exhibits greater phosphorus enrichment compared to the flowing Fu River. Specifically, we observed elevated phosphorus levels in the lake, whereas total nitrogen concentration was significantly higher in the river. Previous studies indicate that the total phosphorus (TP) fluxes from other four rivers connecting to Lake Poyang exceed those from the Fu River (Wang et al., 2024), which may partly explain the lake's notably higher phosphorus concentrations. In the Fu River, the TN concentration has increased gradually due to agricultural pollution in the Gan-Fu Plain irrigation district, combined with industrial and domestic sewage discharges (Zhang et al., 2024a). The conversion of cropland to urban construction land has further intensified nitrogen pollution (Wang et al., 2025). The contrasting nutrient enrichment (e.g., phosphorus in the lake and nitrogen in the river) drives divergent resource use efficiency (RUE) patterns between ecosystems. Phytoplankton communities dominated by Bacillariophyta in the river and Cyanobacteria in the lake exhibit distinct nitrogen-phosphorus competition strategies (Paerl et al., 2016; Schindler et al., 2016), which in turn affect the grazing efficiency of zooplankton (Li et al., 2020a; Liu et al., 2022), particularly for planktonic crustaceans (Cladocera and Copepoda) (Matsuzaki et al., 2018).

Key drivers of resource use efficiency in rivers and lakes

The linear mixed-effects models, accounting physical and chemical parameters as well as seasonal variations, revealed that eutrophication is one of the key drivers of phytoplankton resource use efficiency both in rivers and lakes. However, in the Fu River, eutrophication (high TLI) weakened RUE_{pp} and RUE_{pn}, which further corroborates the adverse impact of eutrophication on the ecological conditions (Zhang et al., 2016). In Poyang Lake, our analysis revealed significant negative effects of PO_4-P and NO_3-N on RUE_{pp} and RUE_{pn}, respectively. Similar studies have also highlighted that nutrient availability is a primary determinant influencing primary productivity, community stability, and biodiversity (Yue et al., 2019). The processes of nutrient absorption and metabolism are regulated by plankton body size and trophic interactions (Hart & Bychek, 2011; Ward et al., 2017), where larger zooplankton (e.g., *Diaphanosoma*) have higher grazing efficiency on ultraplankton including Cyanobacteria (*Synechococcus*) and Chlorophyta (*Nannochloris*) (Bogdan & Gilbert, 1984). Under eutrophication stress and high grazing pressure, many phytoplankton species have evolved or more actively implemented defense strategies (Li et al., 2024), potentially explaining the weak correlation between phytoplankton RUE and nutrient factors in the lake. However, traits-mediated complex regulatory mechanisms between the environment and organisms still require in-depth analysis through additional case studies or controlled experiments.

From our research, the fitting of linear mixed-effects model (measured by R^2) is significantly improved when different hydrological seasons are considered as random effects compared to models that do not account for these effects. Likelihood ratio test results showed that seasonal effects significantly

influenced RUE_{pp} of the Fu River ($\chi^2 = 9.51$, $p = 0.002$), both RUE_{pp} ($\chi^2 = 11.82$, $p = 0.0006$) and RUE_{pn} ($\chi^2 = 11.31$, $p = 0.0008$) of Poyang Lake. However, seasonal effects had no significant impact on RUE_{zp} (Table S6). These clearly demonstrate that the influence of seasonal variation cannot be overlooked. It is lined with other studies from Xu et al. (2023) and Zhang et al. (2024a, b). For example, Zhang et al. (2024a, b) found that the cyanobacteria dominance and RUE_{pn} fluctuated most significantly in spring, and the changes of nutrition and environment conditions in spring significantly has a substantial impact on the growth and reproduction of cyanobacteria, thereby heightening the likelihood of bloom formation.

We hypothesize that seasonal water level fluctuations drive significant changes in environmental factors and plankton populations, thereby influencing RUE trends. This aligns with the pronounced seasonal differences in water levels observed in the Fu River and Poyang Lake. Higher water levels tend to dilute nitrogen, phosphorus, and other pollutants in the water, counteracting the nutrient increases caused by rising water levels (Liu et al., 2016). For instance, research has demonstrated a notable inverse relationship between water levels and the concentrations of nitrogen, phosphorus, and other pollutants in Poyang Lake (Liu et al., 2016, 2020b). An assessment of nutrient concentrations in more than thirty shallow lakes in Yangtze River basin indicates that ammonia–nitrogen, nitrate-nitrogen, and total nitrogen during dry season are 2–3 times higher than during wet season. These changes were associated with increases in total phosphorus concentrations and water temperature (Wu et al., 2006). The increasing nutrient load led to significant shifts in phytoplankton communities from oligotrophic to mesotrophic conditions (Luong et al., 2024).

Conclusion

By characterizing the relationship between resource use efficiency of plankton communities and environmental variables on the river–lake continuum, we found that the RUE of Fu River and Poyang Lake ecosystems were significantly different among seasons and space. The RUE of phytoplankton was markedly exceeded that observed in the lake, while the RUE of zooplankton was higher than that of the river. In addition, RUE is higher in normal and wet season compared to dry season. Our results show that eutrophication is the main driver of phytoplankton RUE in the river and the lake, while RUE_{zp} in river and lake systems is mainly driven by physical and chemical parameters. As the biodiversity-ecosystem function relationship in the river–lake continuum is unclear, this study offers valuable perspectives on how the environmental conditions influence the structure and function dynamics of plankton communities in freshwater ecosystem.

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Data availability The dataset generated and analyzed in the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare they have no conflict of interest. Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

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